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tuxalin and alinloghin updated preview image 5370b57 on Aug 15, 2019 16 commits

screenshots	updated preview image	3 years ago
scripts	some code cleanup	4 years ago
shaders	some code cleanup	4 years ago
textures	initial release	4 years ago
.gitattributes	Initial commit	4 years ago
LICENSE	Initial commit	4 years ago
README.md	updated preview image	3 years ago

About

Procedural water shader for general use.

reflection shader unity3d procedural hlsl water foam displacement

Readme
MIT License
212 stars
13 watching
21 forks

Releases

No releases published

Packages

No packages published

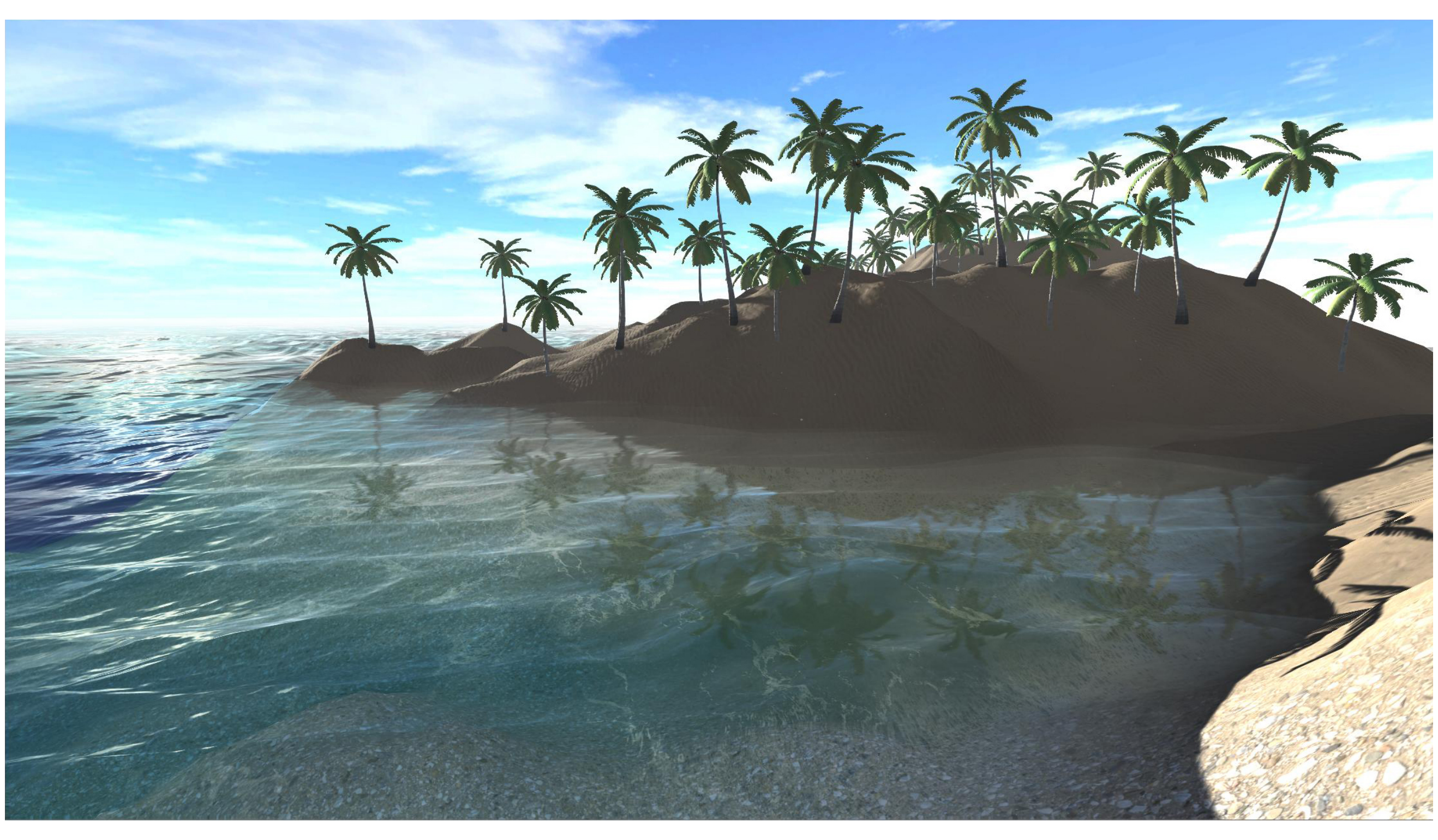
Languages

HLSL 67.6% ShaderLab 27.5%
C# 4.2% Batchfile 0.7%

README.md

Procedural Water

A procedural water shader, multi platforrm support(Unity, GLSL, HLSL), for a variety of water types.



Features

The water shader is designed for various water types: lakes, ocean and rivers.

Current implemented features are:

- multiple maps(height, normal, shore, sky)
- configurable refraction and reflection
- colour extiction based water depth
- mean sky radiance
- displacement
- enabling/disabling features
- bicubic filtering
- configurable shore foam
- HLSL/GLSL/Unity support(for GLSL use the conversion script to SPIR-V)

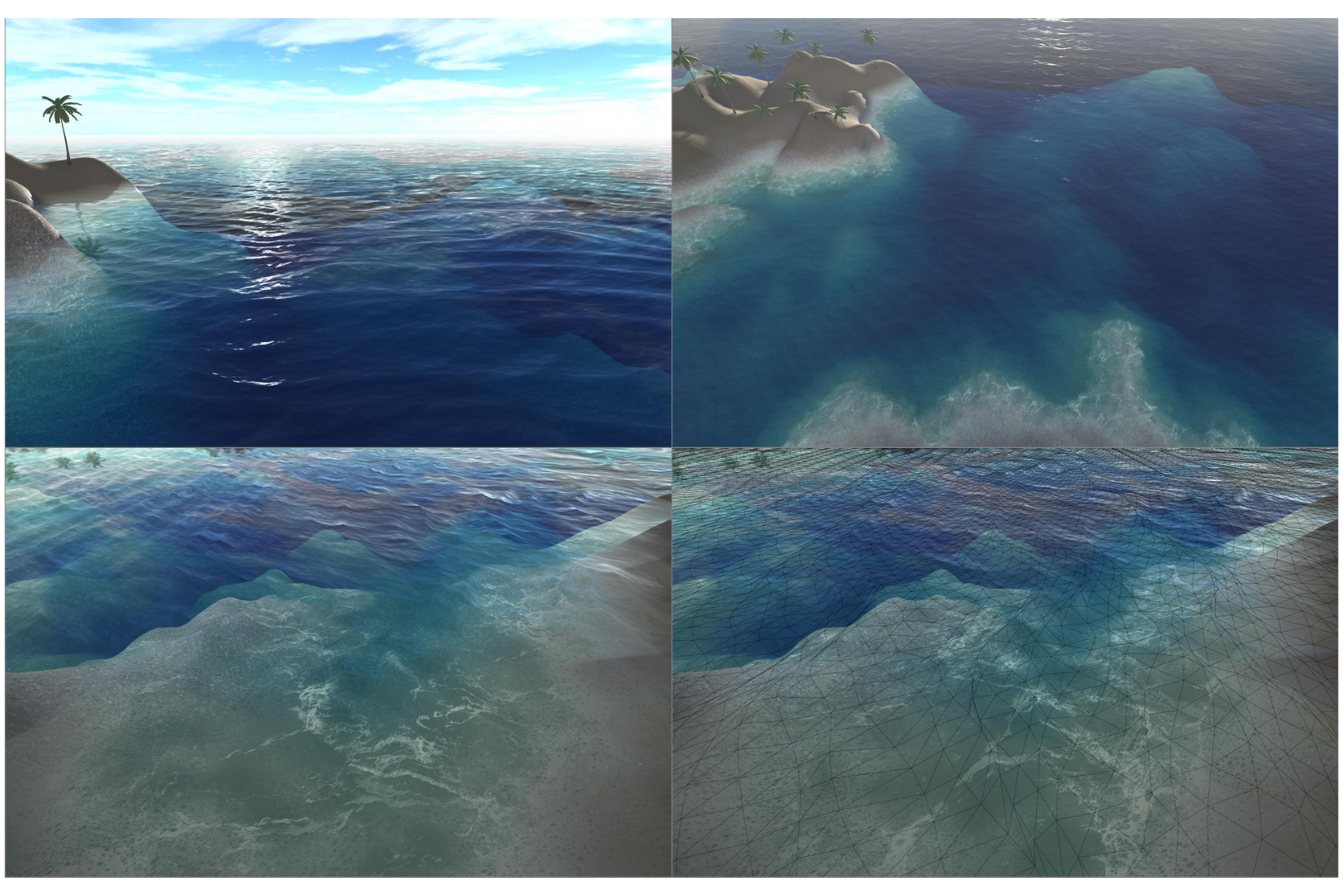
Usage

The shader requires refraction and reflection maps of the scene, the refraction map also requires the depth. See the textures folder for examples of sky, shore, foam, height and normal maps.

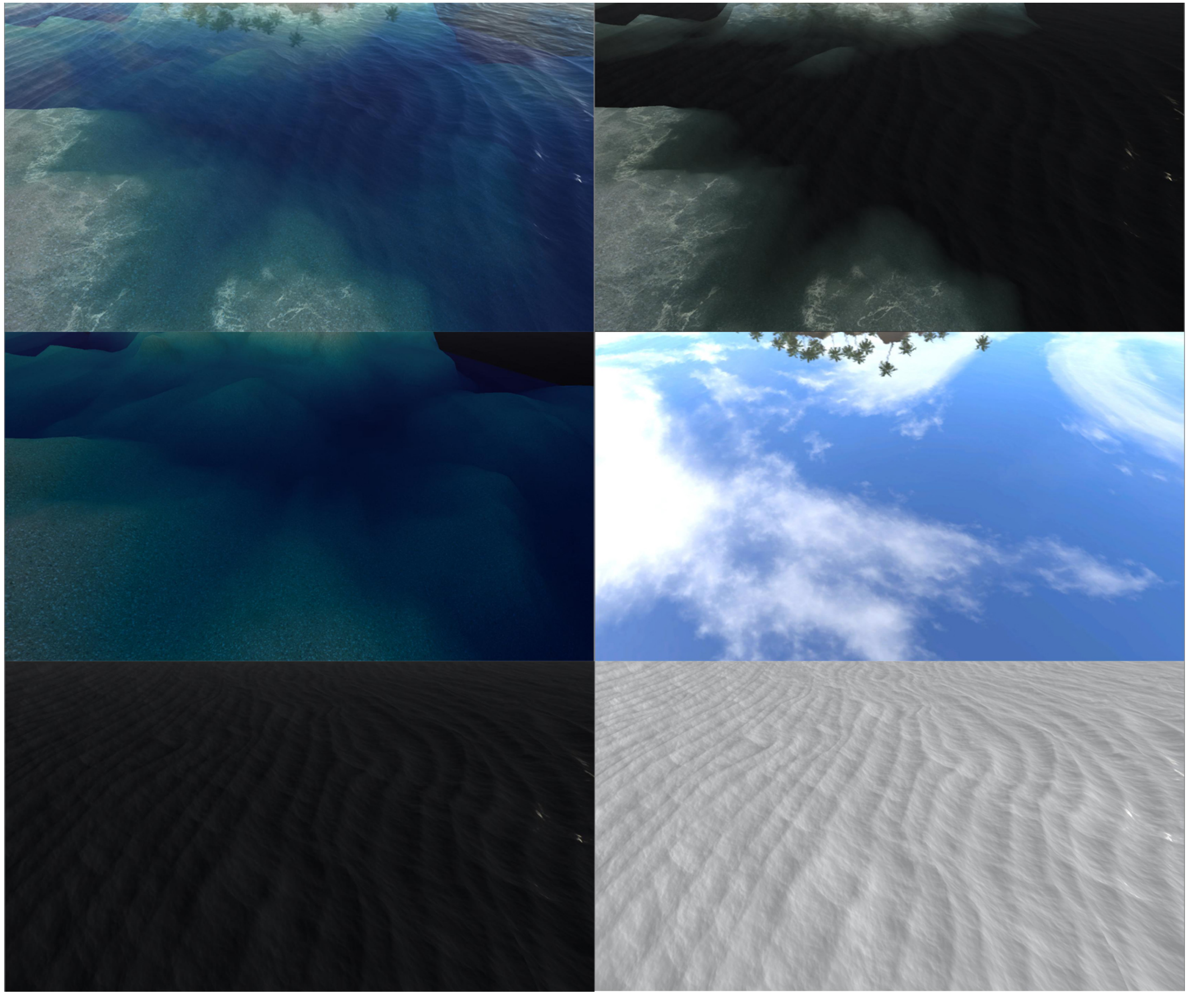
Certain features can be switched off via the following defines:

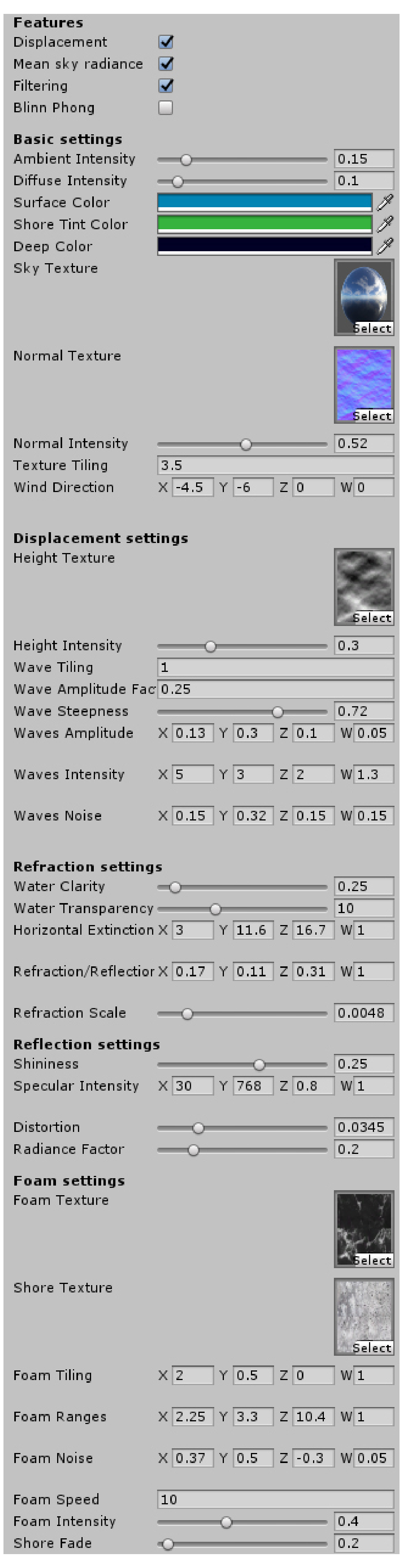
- USE_DISPLACEMENT, adds displacement of gerstner and sinve waves, with an additional height map
- USE_MEAN_SKY_RADIANCE, adds mean sky radiance reflectance, requires a sky map
- USE_FILTERING, enable/disable bicubic filtering
- BLINN_PHONG, use simplified specular

Preview



Composition of the water(final, shore, refraction+depth colour, reflection + distortion, diffuse + ambient, normals):





Features
Displacement ☑
Mean sky radiance ☑
Filtering ☑
Blinn Phong ☐

Basic settings
Ambient Intensity 0.15
Diffuse Intensity 0.1
Surface Color
Shore Tint Color
Deep Color
Sky Texture
Normal Texture
Normal Intensity 0.52
Texture Tiling 3.5
Wind Direction X -4.5 Y -6 Z 0 W 0

Displacement settings
Height Texture
Height Intensity 0.3
Wave Tiling 1
Wave Amplitude Fac 0.25
Wave Steepness 0.72
Waves Amplitude X 0.13 Y 0.3 Z 0.1 W 0.05
Waves Intensity X 5 Y 3 Z 2 W 1.3
Waves Noise X 0.15 Y 0.32 Z 0.15 W 0.15

Refraction settings
Water Clarity 0.25
Water Transparency 10
Horizontal Extinction X 3 Y 11.6 Z 16.7 W 1
Refraction/Reflectior X 0.17 Y 0.11 Z 0.31 W 1
Refraction Scale 0.0048

Reflection settings
Shininess 0.25
Specular Intensity X 30 Y 768 Z 0.8 W 1
Distortion 0.0345
Radiance Factor 0.2

Foam settings
Foam Texture
Shore Texture
Foam Tiling X 2 Y 0.5 Z 0 W 1
Foam Ranges X 2.25 Y 3.3 Z 10.4 W 1
Foam Noise X 0.37 Y 0.5 Z -0.3 W 0.05
Foam Speed 10
Foam Intensity 0.4
Shore Fade 0.2

Future improvements

Planned improvements are:

- caustics
- underwater mode with light rays
- cascading depth maps
- improved water displacement/simulation(FFT)

Contributing

Based on:

- Rendering Water as a Post-process Effect by Wojciech Toman.
- GPU Gems Chapter 1. Effective Water Simulation from Physical Models, by Mark Finch and Cyan Worlds
- Real-time Realistic Ocean Lighting using Seamless Transitions from Geometry to BRDF by Eric Bruneton, Fabrice Neyret, Nicolas Holzschuch
- HLSL 2D simplex noise function by Ian McEwan, Ashima Arts

Bug reports and pull requests are welcome on GitHub at https://github.com/tuxalin/water-shader.

License

The code is available as open source under the terms of the MIT License.

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